

# Java

```
class Demo{

    int a;
    void display(){ // method
        System.out.println("hello");
    }
}

public class App {
    public static void main(String[] args) throws Exception {

        int a = 10; // 4

        double b = a; // 8 // implicit type conversation

        // left variable should be high in size

        double c = 10;

        String result = a > c ? "True" : "False";

        System.out.println(result);

        int d = (int) c; // explicit

        System.out.println(d);
        System.out.println(b);

        System.out.println("Java application");
    }
}
```

```
import java.util.Scanner;

public class Day2 {

    public static void main(String[] args) {

        // int a=10;
        // int b = 20;
```

```
// String result = (a>b) ? "A is greater" : "B is greater";

// System.out.println(result);
// System.out.println(result);

Scanner inp = new Scanner(System.in);

// System.out.print("Enter your number:");
// int c = inp.nextInt();

// System.out.println(c);

// System.out.print("Enter your Float:");

// float marks = inp.nextFloat();

// System.out.println(marks);

// System.out.print("Double :");

// double PI = inp.nextDouble();

// System.out.println(PI);

// System.out.print("Enter your name:");

// // String name = inp.next();
// String name = inp.nextLine();
// System.out.println(name);

// System.out.print("Enter Grade: ");
// char grade = inp.next().charAt(0);
// System.out.println(grade);

// boolean isEligible = inp.nextBoolean();

// System.out.println(isEligible);

int arr[] = {70,45,67,89,30,90};

int max = 0;
for(int i=0; i<arr.length; i++){

    if(arr[i] > max ){
        max = arr[i];
    }
}
```

```

        // System.out.println(arr[i]);
    }

    System.out.println(max);

    int arr2d[][] = { {1,2,3,4}, {4,5,6,4}, {7,8,9,4} };

    for(int i=0; i<arr2d.length; i++){

        // column

        for(int j=0; j<arr2d[0].length; j++){

            System.out.print(arr2d[i][j] + " " );
        }
        System.out.println();
    }

    // jagged array
    int jagged[][] = { {1,2,3}, {4,4}, {7,8,9,4, 8, 9} };

    for(int i=0; i<jagged.length; i++){

        // column
        for(int j=0; j<jagged[i].length; j++){

            System.out.print(jagged[i][j] + " " );
        }
        System.out.println();
    }

}
}

```

```

public class Day3 {

    public static void main(String[] args) {

```

```

// System.out.println("Linear search");

String name = "FIIT"; // string literal
// SCP => String constant pool
String name1 = "fiit";

// String name1 = new String("FIIT");
// System.out.println(name == name1);

// System.out.println(name.equals(name1));

// string method


System.out.println(name.length());
System.out.println(name.toUpperCase());
System.out.println(name.toLowerCase());
System.out.println(name.charAt(3));

System.out.println(name.indexOf("I"));
System.out.println(name.lastIndexOf("I"));

System.out.println(name);
System.out.println(name.trim());

System.out.println(name.equalsIgnoreCase(name1));

System.out.println(name.contains("Tuu")); // returns true or false

System.out.println(name.substring(0, 2)); // endIndex n-1 -> 2-1 = 1

System.out.println(name.replace("F", "f"));

System.out.println(name.startsWith("TFI"));
System.out.println(name.endsWith("T"));

System.out.println(name.concat(" Academy"));
// int arr[] = {10,40,30,56,79,98,100,340};

// int max =0 ;
// for(int i=0; i<arr.length; i++){

//     if(arr[i] > max){
//         // 10 > 0
//         max = arr[i];
//     }

```

```

    // }

    // System.out.println(max);

}

}

```

## Theory

- **Type Conversion**
  - **Implicit (Widening)** → smaller type automatically converted to larger type (e.g., int → double).
  - **Explicit (Narrowing)** → larger type manually cast to smaller type (e.g., double → int).
- **Ternary Operator**
  - Shorthand for if-else.
  - Syntax: condition ? value\_if\_true : value\_if\_false.
- **Scanner Class**
  - Used for input in Java (nextInt(), nextFloat(), nextLine(), nextBoolean()).
- **Arrays**
  - **1D array** → int arr[] = {10,20,30};.
  - **2D array** → int arr2d[][] = { {1,2}, {3,4} };
  - **Jagged array** → arrays with different column sizes.
- **Strings**
  - Immutable objects stored in **String Constant Pool (SCP)**.
  - Common methods: length(), toUpperCase(), toLowerCase(), charAt(), indexOf(), equals(), equalsIgnoreCase(), substring(), replace(), startsWith(), endsWith(), concat().



## Coding Round Questions

- Type conversion → Write a program to demonstrate implicit and explicit type conversion between int and double.

- Ternary operator → Use the ternary operator to find the maximum of two numbers.
- Scanner input → Write a program that takes user input for name, age, and marks using Scanner and prints them.
- Array max → Find the maximum element in an integer array.
- 2D array → Print all elements of a 2D array in matrix format.
- Jagged array → Create a jagged array and print its elements.
- String methods → Demonstrate at least 5 string methods (length, toUpperCase, substring, replace, equalsIgnoreCase).

### **Machine Coding Round Challenge**

#### **Problem: Student Information System**

##### **Requirements:**

1. Create a Student class with fields: id, name, marks.
2. Use Scanner to input details for 5 students.
3. Store them in an array.
4. Find the student with the highest marks.
5. Print all student details.
6. Demonstrate string methods on student names (e.g., convert to uppercase, check if name starts with a letter).